J1
TOP CPU SOCKET (manual)

J2
MOTHERBOARD PLUG (manual)

48-pin MC68008 pass-through

C23: one shared J1/J2 bypass capacitor

Target 5V / MCU supplies

MCU local decoupling: place at supply pads

Five protected inputs

BAT54X Vf margin accepted; see simulations/bat54/.
MODE_PERMIT: GPIO45 permission input, default passive.
USB_PRESENT_N: GPIO46 active-low VBUS detection.
PSRAM_CS_N: GPIO47 QMI CS1; R4 is a 1k pull-up.

8 MiB QSPI PSRAM

U5 shares QSPI clock/data with the 2 MiB in-package flash.
GPIO47 = QMI CS1; GPIO46 = USB presence.
R4 = 1k CS pull-up: C24/C25 at U5 VDD.
Boot: wait 150us, reset PSRAM, then configure QMI.
EPAD (pad 9) is left unconnected; see hardware notes.

PIO0: GPIO0-31; PIO1: GPIO16-47. Synchronization pending.
J1 pin n = J2 pin n, RESET/HALT: LOW or release only.
Target-only power / TTL bus. Mode interlock remains to be completed.

USB-C (target-powered)

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| RP2354B / 48-pin MC68008 \| NOT FOR FABRICATION | |
| Sheet: / | |
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